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import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.util.ArrayList;

public class GradeAnalyzer {
    static int invalidCount = 0;
    public static void main(String[] args) {
        // Step 1: read scores from file
        String filename = "scores.txt";
        invalidCount = 0;
        ArrayList<Integer> scores = readScores(filename);
        // Step 2: calculate statistics
        double average = calculateAverage(scores);

        // Step 5: Find the Highest and Lowest Scores
        int highest = Integer.MIN_VALUE;
        int lowest = Integer.MAX_VALUE;

        for (int score : scores) {
            if (score > highest) highest = score;
            if (score < lowest) lowest = score;
        }

        // Step 3: write and print report
        writeReport(scores, average, highest, lowest, "report.txt");
    }

    // Step 3: Implement readScores
    // Returns a list of valid scores read from the file
    public static ArrayList<Integer> readScores(String filename) {
        // your code here
        ArrayList<Integer> scores = new ArrayList<>();

        try (BufferedReader reader = new BufferedReader(new
        FileReader(filename))) {
            String line;
            while ((line = reader.readLine()) != null) {
                String s = line.trim();
                if (s.isEmpty()) {
                    continue;
                }
                try {
                    int number = Integer.parseInt(s);
                    scores.add(number);
                } catch (NumberFormatException e) {
                    invalidCount++;
                }
            }
        } catch (IOException e) { // ← Now it's correctly paired with the
try

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        System.out.println("Could not read file: " + e.getMessage());
    }
    return scores;
}

// Step 4: Implement calculateAverage
// Returns the average of a list of scores, or 0.0 if the list is
empty
public static double calculateAverage(ArrayList<Integer> scores) {
    // your code here
    if (scores.isEmpty()) {
        return 0.0;
    }
    double sum = 0;
    for (int n : scores ) sum += n;
    return sum / scores.size();
}

// Step 7: Implement writeReport
// Writes and prints the report
public static void writeReport(ArrayList<Integer> scores,
                                double avg, int high, int low,
                                String outputFile) {
    // your code here
    // Calculate grade bands
    int countA = 0, countB = 0, countC = 0, countD = 0, countF = 0;
    for (int score : scores) {
        if (score >= 90) countA++;
        else if (score >= 80) countB++;
        else if (score >= 70) countC++;
        else if (score >= 60) countD++;
        else countF++;
    }

    try (BufferedWriter writer = new BufferedWriter(new
    FileWriter(outputFile))) {
        String header = "=== GRADE REPORT ===";
        writer.write(header + "\n");
        System.out.println(header);

        writer.write(String.format("Total scores processed: %d\n",
scores.size()));
        System.out.println(String.format("Total scores processed:
%d", scores.size()));
        writer.write(String.format("Invalid lines skipped: %d\n",
invalidCount));
        System.out.println(String.format("Invalid lines skipped: %d",
invalidCount));

        //Grade Stats
        writer.write(String.format("Average score: %.2f\n", avg));
        System.out.println(String.format("Average score: %.2f",
avg));
    }
}

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        writer.write(String.format("Highest score: %d\n", high));
        System.out.println(String.format("Highest score: %d", high));

        writer.write(String.format("Lowest score: %d\n", low));
        System.out.println(String.format("Lowest score: %d", low));

        writer.write(String.format("Total scores: %d\n",
scores.size()));
        System.out.println(String.format("Total scores: %d",
scores.size()));

        //Grade Bands
        writer.write("\nGrade Distribution:\n");
        System.out.println("\nGrade Distribution:");

        writer.write(String.format("A (90-100): %d\n", countA));
        System.out.println(String.format("A (90-100): %d", countA));

        writer.write(String.format("B (80-89): %d\n", countB));
        System.out.println(String.format("B (80-89): %d", countB));

        writer.write(String.format("C (70-79): %d\n", countC));
        System.out.println(String.format("C (70-79): %d", countC));

        writer.write(String.format("D (60-69): %d\n", countD));
        System.out.println(String.format("D (60-69): %d", countD));

        writer.write(String.format("F (0-59): %d\n", countF));
        System.out.println(String.format("F (0-59): %d", countF));

        } catch (IOException e) {
            System.out.println("Could not write report: " +
e.getMessage());
        }

    }
}

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